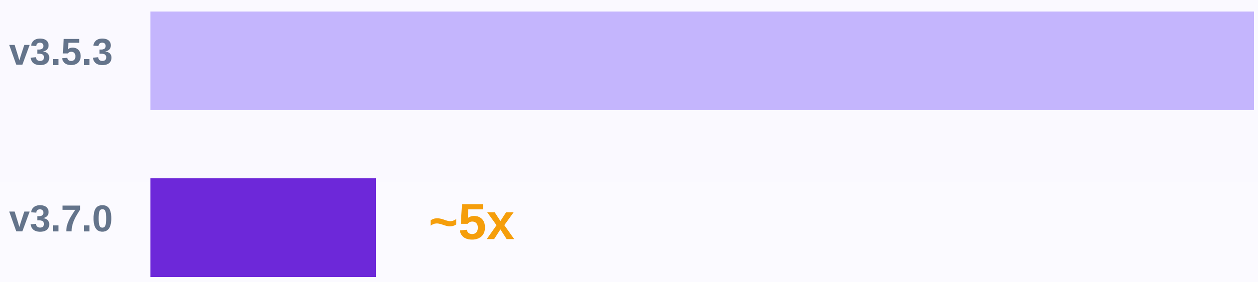
diff - diff

Same estimates.

A fraction of the time.



The v3.5.3 -> v3.7.0 speed & memory arc.

Two-way fixed effects - Callaway-Sant'Anna - Synthetic DiD

THE PROBLEM

Panel scale hits two walls.

Time. And memory.

35,000 store dummies x 3,255,000 rows

= a ~0.9 TB dense design matrix

Fixed-effects absorption never builds that matrix --

but demeaning becomes the hot path: 64-87% of a large-panel fit,

2,948 pandas groupby-transform calls in a single fit.

So we rebuilt the engine.

TWO-WAY FIXED EFFECTS

One engine. Three rebuilds.

pandas transform -> factorize-once bincount MAP -> Rust kernel

93 s -> 20 s

Firm-panel event study - 2.4M rows - 4.6x

2.6 s -> 0.45 s

Geo experiment - 5M rows - 5.8x

now ahead of pyfixest (0.62 s) on this scenario

1.5 s -> 0.61 s

Scanner store-week TWFE - 3.25M rows - 2.5x

at parity with pyfixest

One demeaning engine under TWFE, absorbed DiD, and Sun-Abraham.

MEMORY

Half the high-water mark.

15.3 GB -> 7.8 GB

Peak allocation - 2.4M-row clustered solve

the waste was marshalling, not math -- one owned copy instead of two

~0.9 TB -> 0 bytes

The dummy design matrix we never materialize

Same estimates: the rebuilt engine matches the reference

demeaner to atol 1e-12 on every path.

CALLAWAY-SANT'ANNA

The O(n_units) rewrite.

per-cohort DataFrame scans -> closed-form influence functions

1.32 s -> 0.21 s

Analytical fit - 2M rows - 6.3x

960 MB -> 582 MB

Peak RSS on that same fit

11.6 GB -> 2.4 GB

Multiplier bootstrap at scale - now tiled

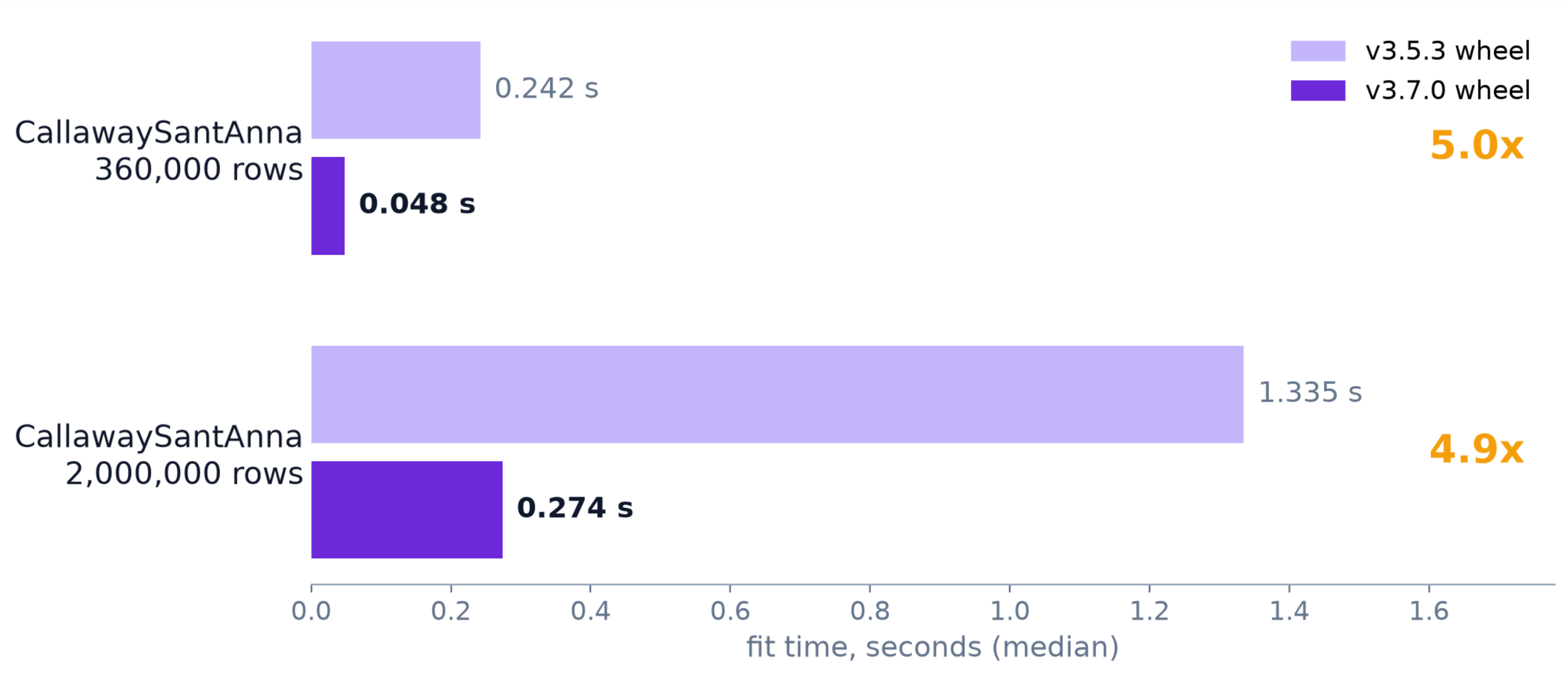
Point estimates bit-identical.

Aggregated SEs within 5e-16 relative of the original path.

THE UPGRADE

Wheel to wheel.

Released PyPI wheels, same data, same machine, fresh processes.



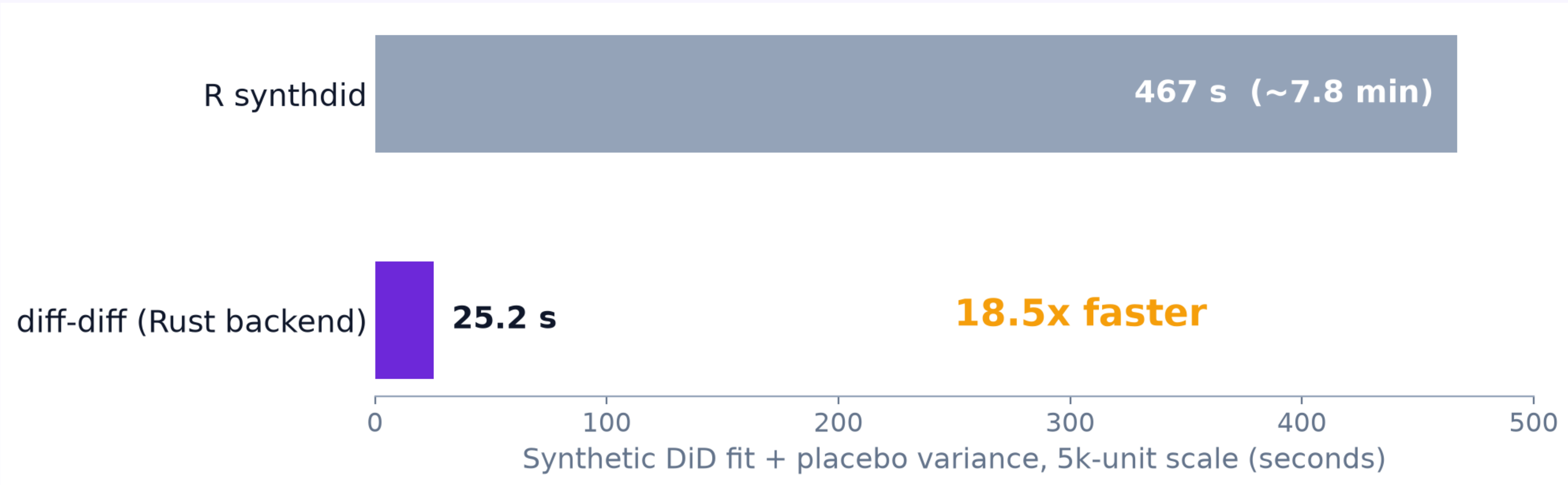
Identical estimates. The upgrade is the migration.

`pip install --upgrade diff-diff`

SYNTHETIC DID

The Rust backend,

re-benchmarked.



18-55x faster than R synthdid across benchmark scales,

at matched 200-replication placebo variance.

Same Frank-Wolfe optimizer: id-aligned weights reproduce R (max diff 3.5e-12).

GATED

Speed that can't drift.

Published benchmarks pass fail-closed parity gates vs R.

Internal rewrites verify against reference outputs -- same estimates.

ATT parity: fail-closed

Every published benchmark gated against R -- a hard failure blocks publication

Effect surfaces: < 1e-6

Per-(g,t), event-study, and group aggregations -- observed max ~5e-11

SDID weights: < 1e-8

Unit and time weights id-aligned to R's ordering -- observed max 3.5e-12

Reproducible end to end

Committed artifacts + one command regenerate every published number

IN PRACTICE

Nothing to change.

Rust bundled in wheels

pip wheels ship the backend --

no toolchain, used automatically

Same sklearn-style API

fit() unchanged -- zero code

changes to pick up the speed

Memory tiling at scale

bootstrap + solver paths sized

for multi-million-row panels

R parity, maintained

benchmarks gate against R --

rewrites verified vs reference

All numbers: Apple M4 Max, medians of repeated runs,

committed benchmark artifacts in the repo.

Faster by default.

v3.7.0.

```
$ pip install --upgrade diff-diff
```

github.com/igerber/diff-diff

diff - diff

Difference-in-Differences for Python